

SMART INDIA HACKATHON 2025

SMARTWORK 360

AI + Blockchain based Smart Task & Performance Management for Government Offices — built as a module on the e-Office platform.



PS ID

SIH 25250

THEME

Miscellaneous

CATEGORY

Software

TEAM ID

95554

TEAM NAME

TECH TRACKERS25

PROBLEM STATEMENT · SIH 25250

Government offices often suffer from low levels of productivity at the organizational, team and individual levels, but there are no online tools to set goals / objectives, define and benchmark success parameters, track progress and calculate productivity levels at the aforesaid 3 levels. To address this management gap and improve productivity and employee engagement in government offices, a suitable online tool / module should be developed under the e-Office platform.



LIVE PROTOTYPE

smartwork360.vercel.app



SOURCE CODE

github.com/Het161/smartwork360



DEMO VIDEO

[Watch on Google Drive](#)

Working prototype — everything in this deck is running today.

Proposed Solution

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What is SMARTWORK 360?

- An online tool on top of NIC e-Office to set goals, assign tasks with SLAs, track progress live and measure productivity.
- One platform, three role-based dashboards — Admin, Manager, Employee.
- Every action is recorded on a tamper-proof audit chain.

Why it is unique

-  **AI assistant** — answers from your live office data
-  **Sentiment AI (DistilBERT)** — tracks team morale
-  **Burnout prediction** — balances workload early
-  **Fraud detection (IsolationForest)** — flags anomalies
-  **SHA-256 audit chain** — one-click tamper check
-  **English + Hindi UI** — works offline too

Solves the problem statement at all 3 levels



ORGANISATION → **Admin Dashboard**

Org-wide KPIs · department comparison · full audit explorer



TEAM → **Manager Dashboard**

Team goals · SLA tracking · workload, morale & burnout view



INDIVIDUAL → **Employee Dashboard**

My tasks & progress · personal performance score · AI assistant

30–40%

Faster workflows

92%

Fraud precision

85%+

Sentiment accuracy

100%

Tamper-evident audit

30+

REST APIs (live)

Technical Approach

Technology Stack

Frontend

Next.js 14 (React) · Tailwind · PWA — web + mobile

Backend

Node.js · Express · Prisma ORM

Database

PostgreSQL

AI / ML

Python FastAPI · DistilBERT · Scikit-learn

Blockchain

SHA-256 hash-chain (ETH / Hyperledger planned)

Security

JWT + RBAC · Parichay SSO (sandbox)

APIs & Cloud

30+ REST APIs (Swagger) · Vercel · NIC / AWS-ready

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 Admin

 Manager

 Employee

Next.js Web App (PWA) — role-based dashboards · English + Hindi · WCAG 2.1 AA

Express REST API — JWT + RBAC · 30+ documented endpoints · SLA cron alerts

PostgreSQL

Prisma ORM · realistic seeded demo data

FastAPI ML Service

sentiment · burnout · fraud · AI assistant (offline-safe)

SHA-256 Audit Chain

every action hashed & linked in sequence

Tamper-proof audit trail (blockchain-style)



Change any old record → hashes break → tampering is detected instantly (live demo in the prototype).

Feasibility & Viability

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Feasibility

- Working prototype is live today — smartwork360.vercel.app
- Open-source stack → very low cost
- One PWA runs on desktop and mobile
- Deployable on NIC Cloud / AWS / Vercel
- GIGW 3.0-aware design · WCAG 2.1 AA accessible



Challenges & Risks

- Staff used to manual files may resist change
- Strict government data-privacy rules
- Integration with legacy e-Office modules
- Patchy internet in field offices
- Initial training and onboarding effort



Our Strategies

- Stepwise rollout: pilot → feedback → launch
- Built-in AI assistant trains users in-app
- Offline heuristic mode — demo never breaks
- NIC-compliant security + tamper-proof audit
- Role-based access limits data exposure

Rollout plan

Pilot in 1 section



2-week feedback loop



Full department rollout



Scale: Govt · PSUs · Smart Missions

Impact & Benefits

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Potential Impact

- Manual offices become digital and data-driven
- Accountability at organisation, team and individual level
- Faster approvals → better citizen service
- Healthier, motivated workforce (burnout care)
- Bridges field staff and senior officers

Social Benefits



Trust and transparency in governance · better employee morale and recognition · faster service delivery to citizens.

Economic Benefits



Saves time, paper and manpower cost · fewer delays and rework losses · open-source stack means near-zero licence cost.

Environmental Benefits



Paperless, eco-friendly workflow · less printing and travel · lower carbon footprint for every office.

Aligned with

Digital India

Green Governance

NIC e-Office ecosystem

Research & References

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4. **Digital India Programme, MeitY** — framework for digital transformation of public offices
www.digitalindia.gov.in
5. **Sanh et al., “DistilBERT” (2019)** — the sentiment model we use
arxiv.org/abs/1910.01108
6. **Liu, Ting & Zhou, “Isolation Forest”, IEEE ICDM (2008)** — the anomaly / fraud detection algorithm
ieeexplore.ieee.org/document/4781136
7. **NITI Aayog — digital transformation reports** — best practices for government workflow systems
www.niti.gov.in

Our Project — Live & Open



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PROTOTYPE: LIVE

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